Some Key Points

About all these Mass Spec acronyms in 2025

False Discovery Rates FDR

- There are lot’s of FDR’s at every level of analysis!
  - Precursor
  - Peptide
  - Protein
  - Differential Expression

- There is evidence some software's FDR calculations are not entirely valid

False Discovery Rates FDR

How do we know we are correct when an algorithm identifies a peptide from complex mass spec data

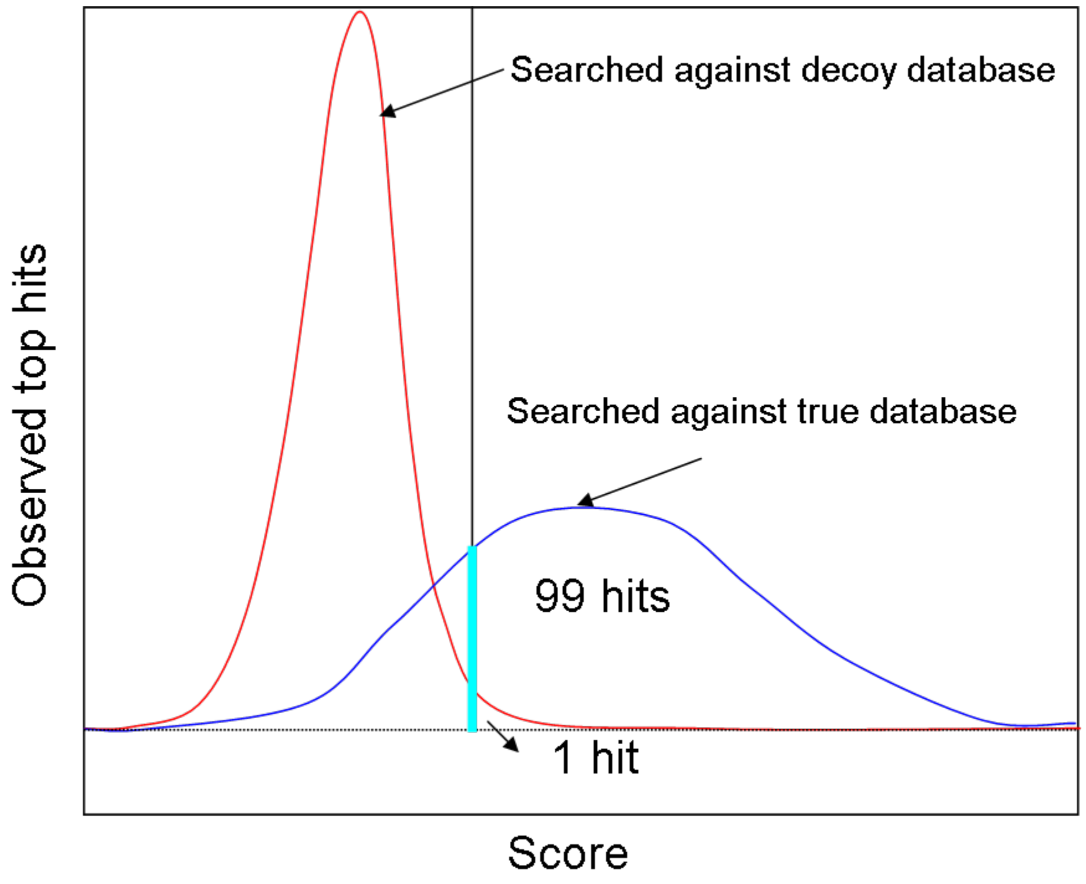
We don’t!!

We use decoys and entrapment data to try and figure this out

What is a Decoy

- A decoy is a fake peptide that is not in your sample
  - Usually these are just reversed peptide sequences
    - PEPTIDE – Real
    - EDITPEP – Decoy

    - This preserves the amino acid proportions in the search



What is an entrapment experiment

The software knows what's a Decoy and what is not

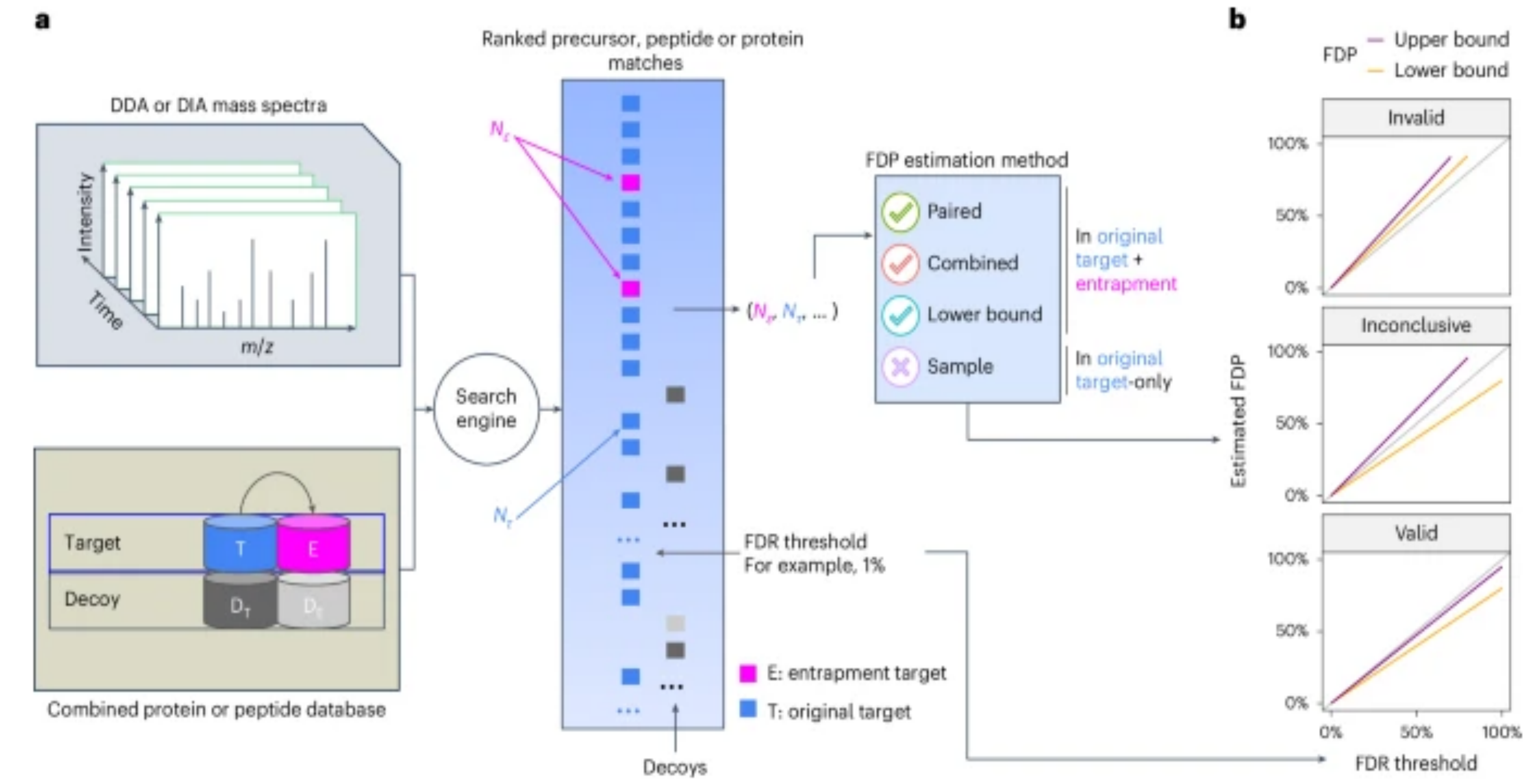
For an entrapment experiment the software does not know what is real and what is not

An example

you have human cells

You search 22K human sequences and 100K Corn sequences

Fig. 1: Entrapment strategy for FDR control evaluation.



You have now entrapped your human sequences with corn sequences

2025 Recent paper you should read

- https://www.nature.com/articles/s41592-025-02719-x

Article | Open access | Published: 16 June 2025

Assessment of false discovery rate control in tandem mass spectrometry analysis using entrapment

Bo Wen, Jack Freestone, Michael Riffle, Michael J. MacCoss, William S. Noble & Uri Keich

Nature Methods **22**, 1454–1463 (2025) | Cite this article

Abstract

A critical challenge in mass spectrometry proteomics is accurately assessing error control, especially given that software tools employ distinct methods for reporting errors. Many tools are closed-source and poorly documented, leading to inconsistent validation strategies. Here we identify three prevalent methods for validating false discovery rate (FDR) control: one invalid, one providing only a lower bound, and one valid but under-powered. The result is that the proteomics community has limited insight into actual FDR control effectiveness, especially for data-independent acquisition (DIA) analyses. We propose a theoretical framework for entrapment experiments, allowing us to rigorously characterize different approaches. Moreover, we introduce a more powerful evaluation method and apply it alongside existing techniques to assess existing tools. We first validate our analysis in the better-understood data-dependent acquisition setup, and then, we analyze DIA data, where we find that no DIA search tool consistently controls the FDR, with particularly poor

Also read

- https://github.com/vdemichev/DiaNN/discussions/1035

FDR in proteomics & data filtering #1035

vdemichev announced in **DIA proteomics: in detail**

vdemichev on Jun 9, 2024 **Maintainer**

This is a note on the concept of false discovery rate (FDR) in proteomics, what do the q-values produced by the software mean, how to (and not to) estimate the 'real FDR' and how to compare different software tools.

The concept of false discoveries in LC-MS proteomics and entrapment validation

First, I would like to discuss the definition of what it means that a particular precursor ion identification is false. One can consider three potential definitions:

- **F1.** Identification is false if none of the signals recorded by the mass spec in the matched spectrum originate from the same precursor ion species as reported by the software.
- **F2.** Identification is false if none of the signals recorded by the mass spec in the matched spectrum originate from the same or 'equivalent' precursor ion species as reported by the software. I will talk about what 'equivalent' means a bit later.
- **F3.** Identification is false if the sample does not contain the reported peptide or an 'equivalent' peptide.

F1 is the strictest definition, and FDR in the sense of F1, as will be clear from below, is almost never well controlled by the modern software. It might seem that F2 and F3 are quite similar, however the false discovery rates according to these definitions can in fact be dramatically

Retention times and how we predict them

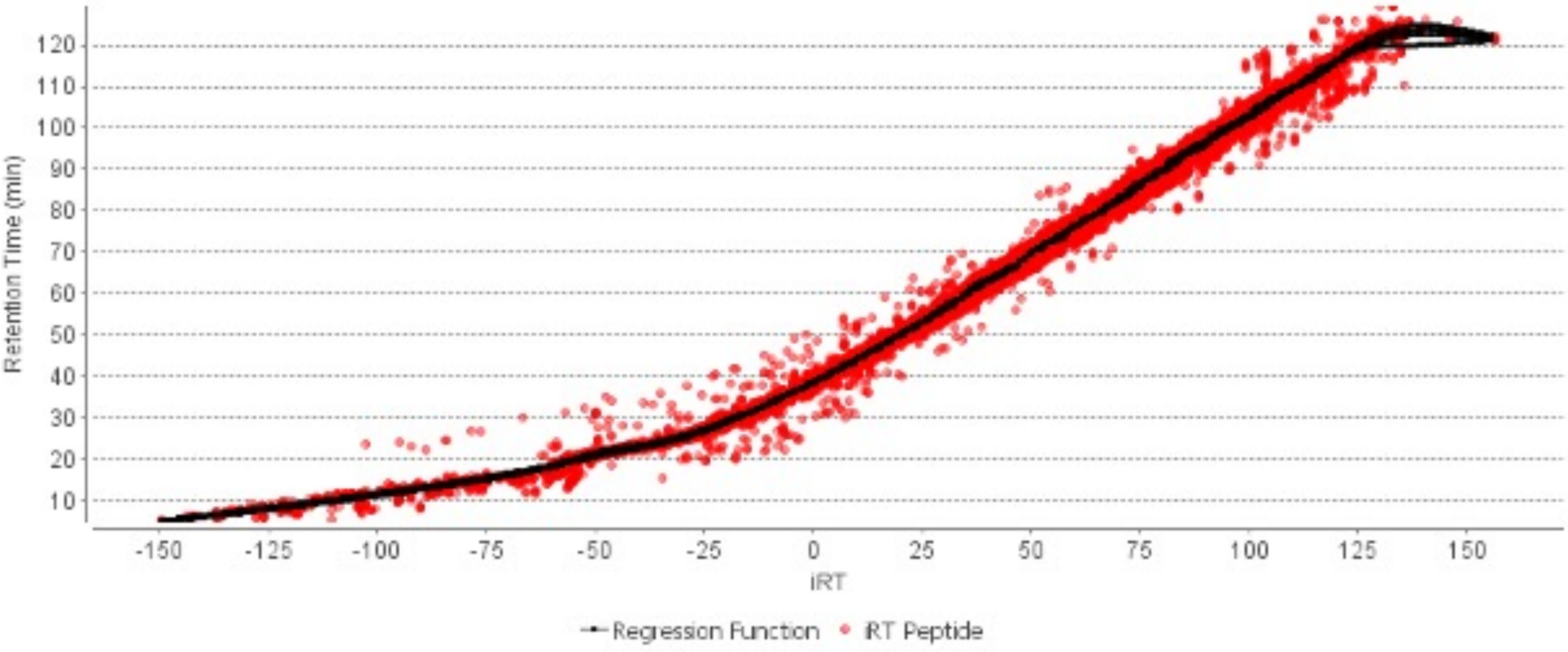
iRT and CIRT’s

We can predict when a peptide will elute from a column within 20-30 seconds

Regardless of the column, analysis length, sample

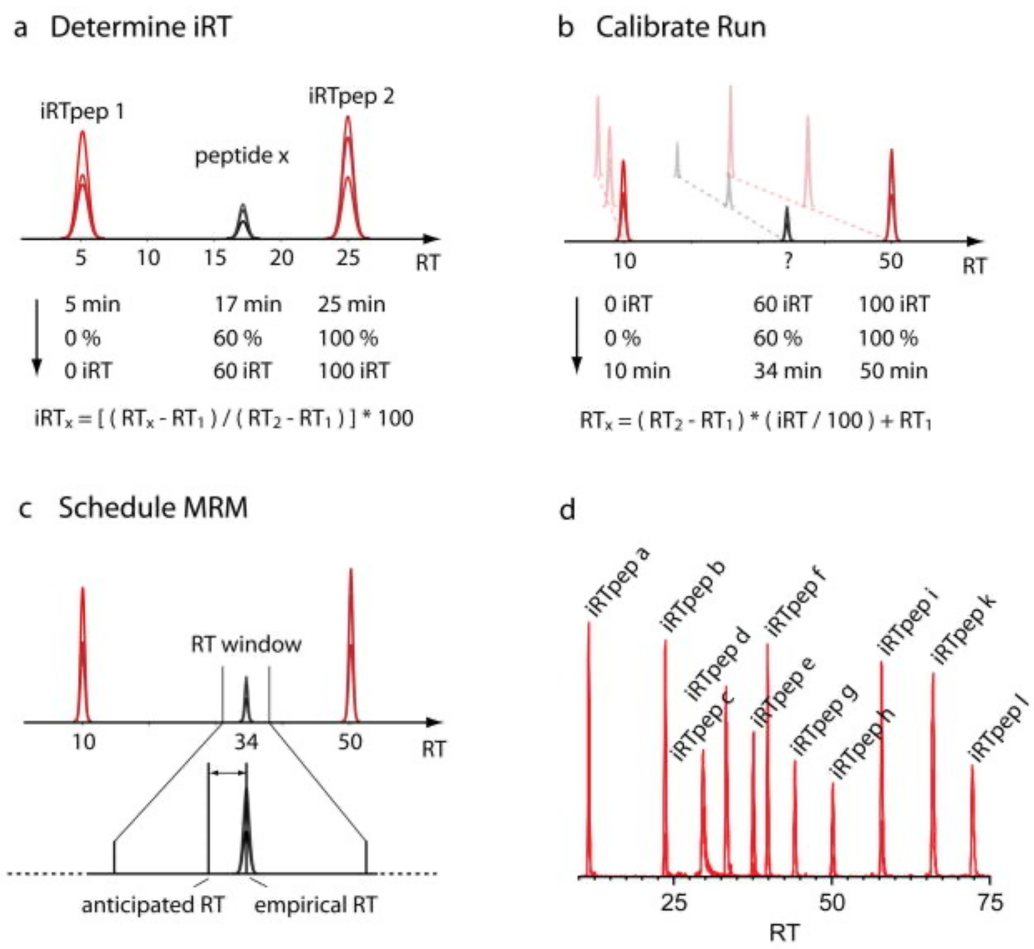
We index them !!

https://pmc.ncbi.nlm.nih.gov/articles/PMC3918884/

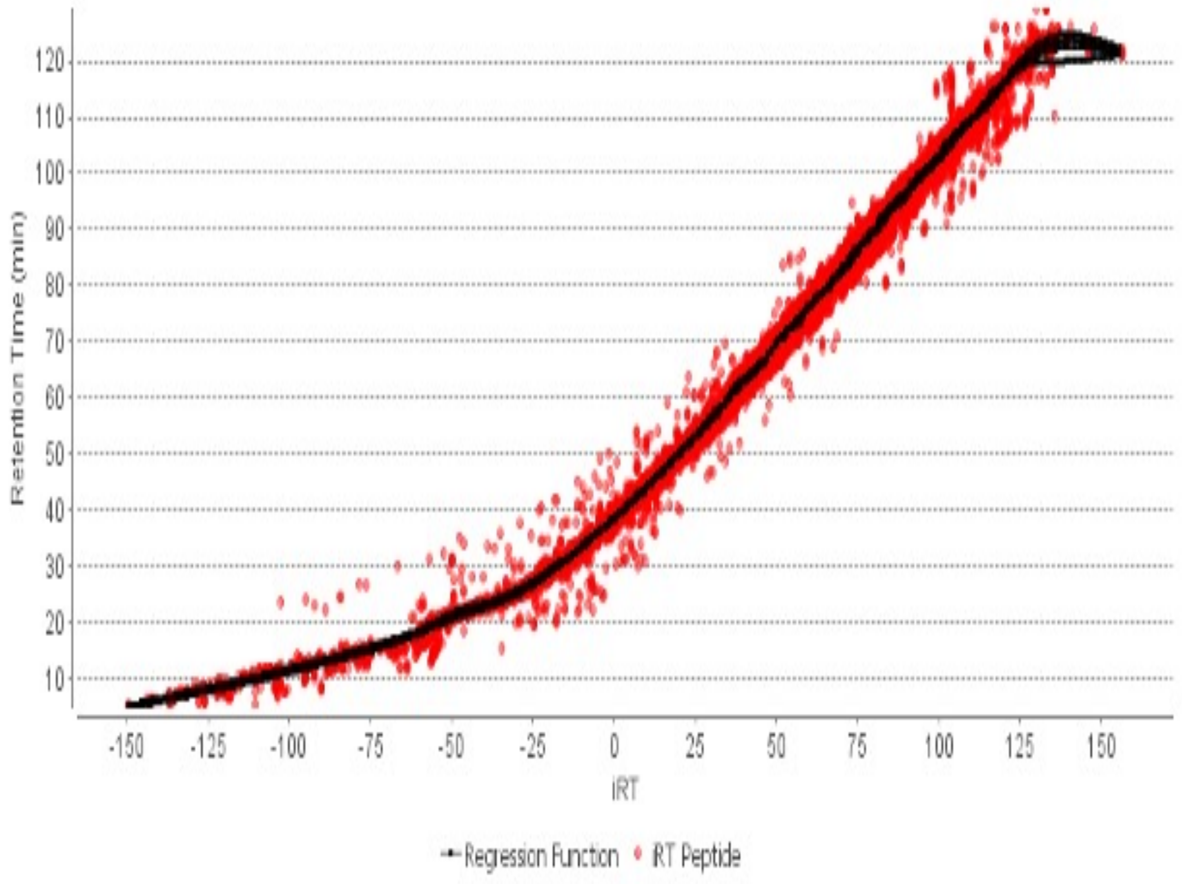


At first people spiked in a set of peptides

https://pmc.ncbi.nlm.nih.gov/articles/PMC3918884/



Plot the iRT space and RT space of these peptides



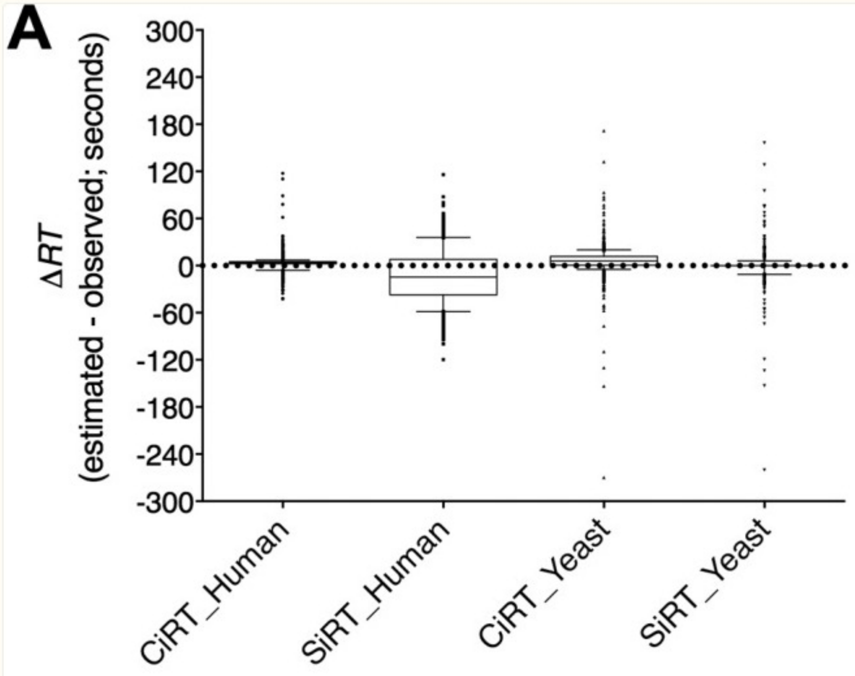
Most hydrophobic = +100 iRT
Most hydrophilic = -25 IRT

This way it doesn't matter what your run time is . You just recalibrate and do a linear regression

Now we do not spike in peptides

- We use endogenous peptides that are already in your sample
  - CiRT
    - https://pmc.ncbi.nlm.nih.gov/articles/PMC4597153/

Fig. 2.



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Identification of a Set of Conserved Eukaryotic Internal Retention Time Standards for Data-independent Acquisition Mass Spectrometry*

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How do we